# Enhanced Volatility-Managed Allocation with GARCH, CVaR, and a Gold Defensive Sleeve

Abstract: This project extends an original SPY/TLT volatility-managed allocation strategy by adding GLD as a third defensive asset. The final model uses monthly walk-forward rebalancing, GARCH(1,1) volatility forecasts, and CVaR-aware optimization to allocate across SPY, TLT, and GLD.

Key idea: The original SPY/TLT proposal was vulnerable when stocks and long-duration Treasuries sold off together. Adding gold improves the robustness of the defensive sleeve and leads to better risk-adjusted performance.

Headline result: The enhanced SPY/TLT/GLD strategy improves Sharpe ratio from 0.889 to 0.991, cuts max drawdown from -28.27% to -22.67%, and lowers monthly CVaR 95% from 5.70% to 4.63% relative to the original dynamic model.

# Methodology and Results

Dataset: Daily adjusted close prices for SPY, TLT, and GLD from 2008-01-01 to 2024-12-31.

Walk-forward design: Monthly rebalancing with a trailing 756-day estimation window.

Risk model: Per-asset GARCH(1,1) one-month volatility forecast with an EWMA fallback.

Allocation rule: Choose weights that maximize expected return minus a CVaR penalty and turnover penalty, with SPY constrained between 10% and 90%.

Performance summary: Enhanced Dynamic SPY/TLT/GLD achieved cumulative return 2.3750, annualized return 9.08%, annualized volatility 9.16%, Sharpe ratio 0.9913, max drawdown -22.67%, and monthly CVaR 95% 4.63%.

Interpretation: The enhanced model does not maximize absolute return versus buy-and-hold SPY, but it delivers the strongest overall risk-adjusted profile among the tested strategies and handles the 2022-2024 regime substantially better than the original design.

# DRL Extension

The repository also includes a PPO-based deep reinforcement learning extension for monthly asset allocation across SPY, TLT, and GLD.

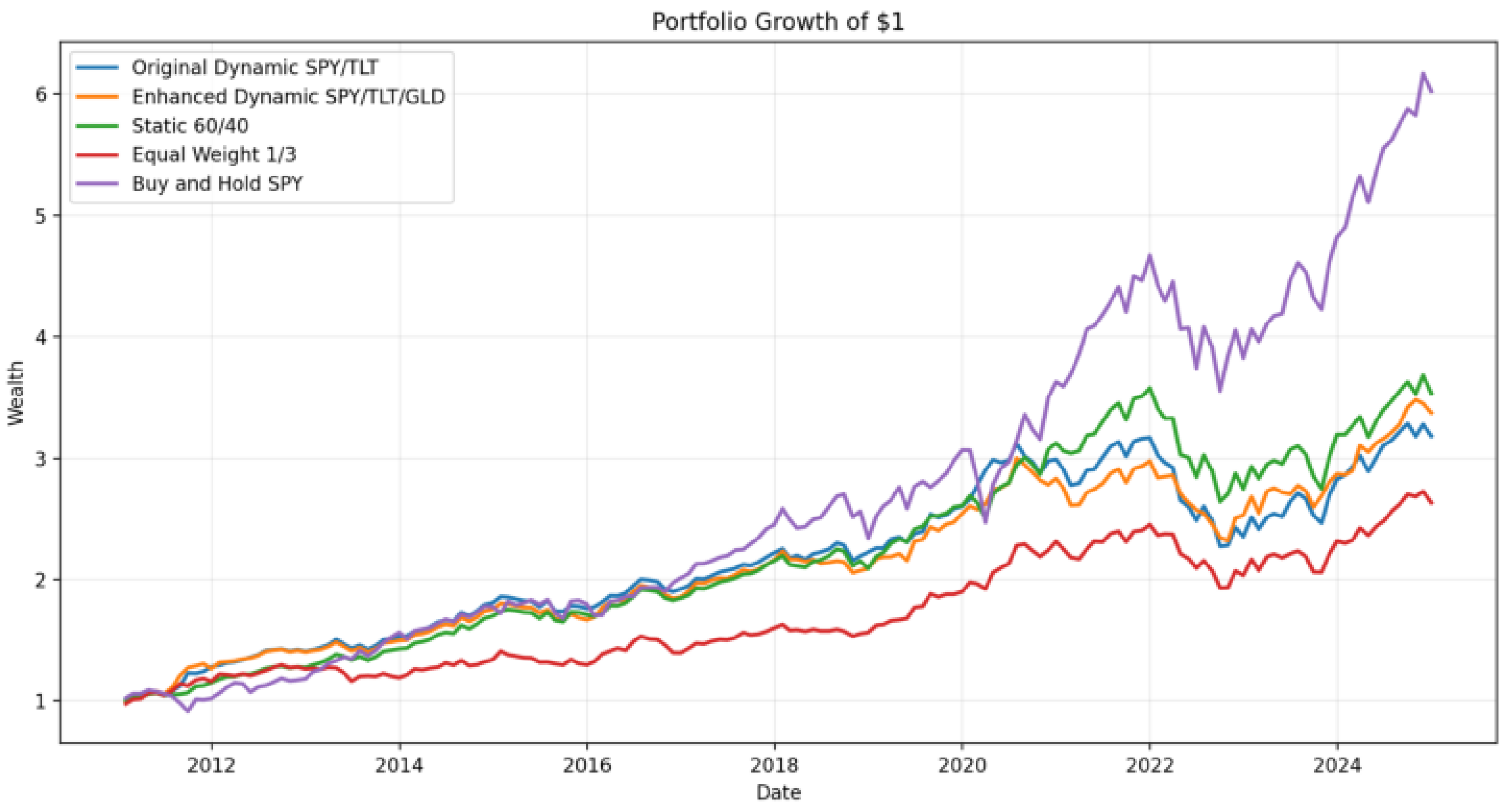
State design: trailing multi-horizon returns, realized volatilities, one-month GARCH forecasts, rolling correlations, and previous portfolio weights.

Action design: a continuous three-asset weight vector normalized to sum to one, with the same long-only and SPY allocation constraints used in the rule-based baseline.

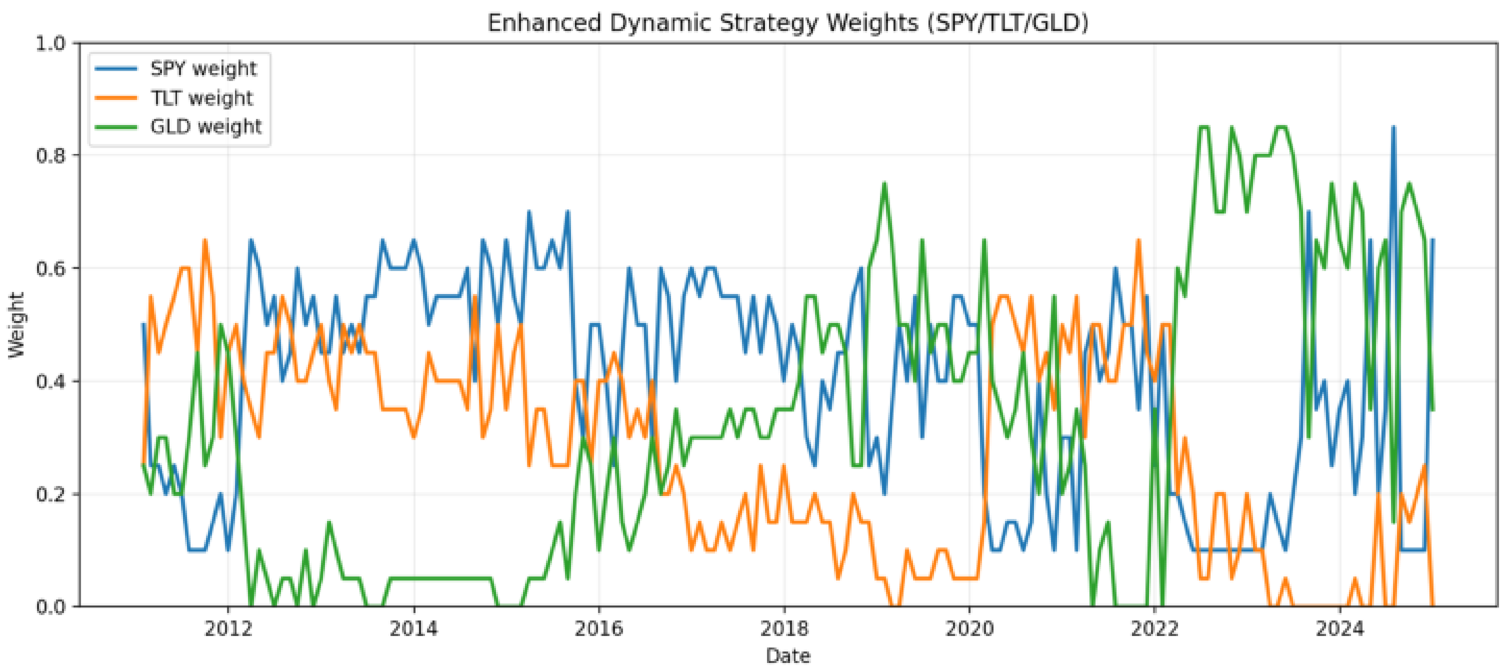
Reward design: realized portfolio return minus volatility and turnover penalties.

Result summary: In the strict 2021-2024 out-of-sample test window, the DRL policy produced stronger return and Sharpe ratio than the rule-based baselines, but its tail risk remained worse than the enhanced GARCH-CVaR allocator. This makes the DRL result promising, but not yet a full replacement for the more interpretable baseline.

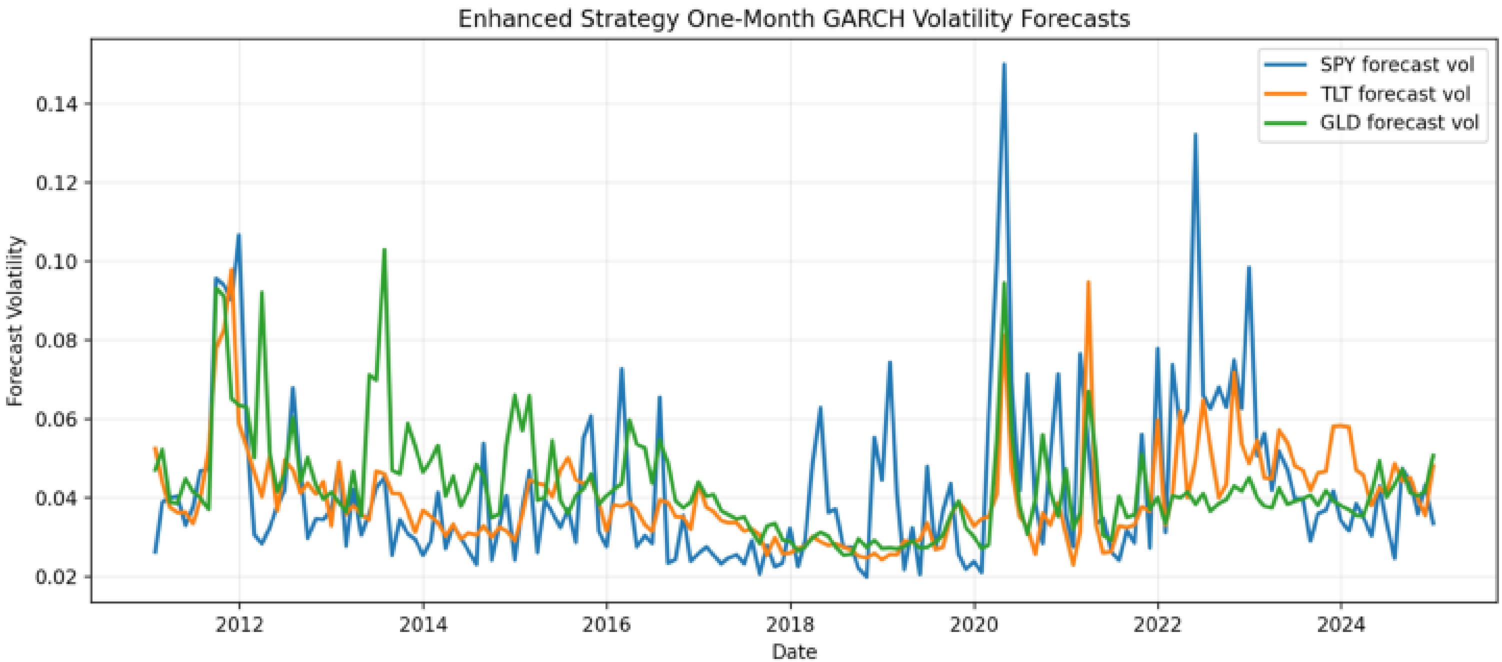
# Equity Curves



# Enhanced Dynamic Weights



# Enhanced Forecast Volatility



# DRL Equity Curves